When Does Privileged Process Supervision Help? A Teacher-Level Sweet Spot, a Verified Transfer Null, and a Supervision-Mismatch Diagnosis

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Abstract

Process reward models (PRMs) score intermediate reasoning steps, and many pipelines construct their step-level supervision automatically from a stronger teacher model. We study two questions. When does privileged (answer-aware) supervision help such a teacher? And can that advantage be distilled into a small, deployable, answer-blind student verifier? On a controlled three-tier difficulty probe (GSM8K, MATH, OlympiadBench), a privileged teacher's step-error localization improves only in a tractability sweet spot. Privilege is redundant on easy problems the teacher can already self-verify (GSM8K, n.s.). It is unusable on the hardest problems, where the reference is too long to exploit (OlympiadBench, n.s.). It helps precisely in between (MATH, +0.051 F1, 95% CI $[0.01, 0.09]$), and the effect replicates across model families (Qwen-27B, +0.082 F1). This advantage does not distill. A 1.5B student trained on the privileged teacher's labels is statistically indistinguishable from one trained on no-GT labels, and neither beats majority vote on downstream best-of-N re-ranking. The null is not explained by pipeline insensitivity. A positive control confirms that the training path tracks teacher quality (+0.050 ROC-AUC, 95% CI $[0.021, 0.078]$, significant). Nor is it explained primarily by student capacity. Under an identical training and evaluation path, a 3B verifier trained on format-matched ProcessBench-style gold labels reaches 0.73–0.79 ROC-AUC across eight seeds on full ProcessBench-MATH, rising to 0.83 with post-hoc score averaging across seeds. The same generated teacher labels remain weak (0.55–0.61 ROC-AUC) at both 1.5B and 3B scale. The evidence points to a supervision distribution and format mismatch, not raw capacity, as the dominant failure mode. It also offers a practical diagnostic. Before scaling students or searching over nearby training recipes, test whether format-matched supervision can support the target verifier at all.

1 Introduction

Process reward models assign scores to intermediate reasoning steps and are increasingly used to supervise or verify mathematical reasoning systems. Because human step-level annotation is expensive, many PRM pipelines construct supervision automatically, either by giving a labeling teacher privileged (answer-aware) access to a reference solution, or by other weak-supervision means. Two questions follow naturally, and most prior work answers only the first: (1) does privileged access actually make the teacher a better labeler, and if so, when; and (2) does that advantage survive distillation into a small, deployable, answer-blind student verifier?

We answer both under a controlled setup that holds the student architecture, optimizer, and evaluation benchmark fixed while varying exactly one thing at a time. First, we characterize *when* privileged supervision helps a teacher (§4): it is not uniformly useful, but pays off only in a tractability window where the teacher both needs the reference (it cannot already self-verify) and can act on it (the reference is short enough to follow). Second, we ask whether this real, mechanistically-understood teacher-level advantage distills into a small student PRM (§5): it does not, and we show this null is not an artifact of an insensitive training pipeline. Third, having ruled out pipeline insensitivity, we ask the natural follow-up in §6. Is the failure to distill a student-capacity problem, or a supervision problem? Under the identical training and evaluation path, we show that format-matched gold process supervision *does* produce a competent verifier, while our generated teacher labels do not, at both 1.5B and 3B student scale. This decomposes what would otherwise be a flat, somewhat unsatisfying null into three connected findings: a real, well-characterized teacher effect;

a verified failure of that effect to distill, with pipeline sensitivity confirmed; and a concrete diagnosis of why it fails, pointing at supervision distribution/format rather than capacity.

The practical payoff is a cost asymmetry. In our setup a single 500-step LoRA diagnostic on format-matched labels costs approximately 0.3 GPU-hours on one GPU, whereas the preliminary generated-label recipe sweep it would have replaced consumed approximately 15 GPU-hours across roughly 50 runs. Both of the results above are ways of paying that smaller bill first: the teacher-level sweet spot says *where* in the difficulty range privileged labeling is worth buying at all, and the format-matched check says *whether* the labels you already have can support the verifier you want. Neither requires training the student you actually intend to ship. We therefore read this work as a data-side contribution to efficient reasoning: a pipeline for deciding, cheaply, whether a process-supervision corpus is worth scaling before any compute is committed to scaling it.

2 Related Work

2.1 Process supervision and process reward models

A line of work has established that supervising *how* a model reasons, step by step, is more effective than supervising only whether the final answer is correct. Lightman et al. [1] compared the two regimes directly and found that process supervision substantially outperforms outcome supervision on the challenging MATH dataset, releasing the large PRM800K corpus of step-level human labels to support the finding. Because human step annotation is expensive, Math-Shepherd [2] showed that process supervision can be constructed automatically, assigning a reward to each solution step without manual labels. The resulting signal was used both to re-rank candidate solutions and to drive step-level reinforcement learning. These process reward models (PRMs) treat each step's correctness as a scalar quantity and deploy the PRM as a frozen scorer or ranker. Our work shares the step-level granularity but departs from the scalar view for its teacher-level analysis: we treat the teacher's step judgment, including its natural-language critique, as the object to be distilled rather than a number to be predicted.

2.2 Generative PRMs

A more recent thread makes the verifier itself generative, producing a reasoning trace before it judges. GenPRM [3] performs explicit chain-of-thought reasoning with code verification before scoring each step, while ThinkPRM [4] fine-tunes a long-CoT verifier that articulates a verification rationale for every step and reaches strong ProcessBench performance from only a few thousand process labels. These models are close to ours in spirit, since they generate critiques rather than emit bare scores. But they are designed as verifiers evaluated at inference time, not as a teacher whose score-plus-critique behavior is distilled into a separate, smaller, answer-blind student. None of them isolates when privileged (answer-aware) judgment actually helps, which is the question we take up in §4.

2.3 Weak and generated process supervision

Recent weak-supervision approaches attempt to train PRMs without dense human process labels, using outcome labels, generated rationales, or model-derived signals [10]. These methods make PRM training more scalable, but they also raise the question we study in §6: when generated labels fail, is the problem the student, the recipe, or the supervision distribution? Noise-aware PRM training directly tackles noisy process supervision through label correction and iterative denoising [11]; our controlled gold-label comparison is complementary because it first asks whether the current generated-label distribution supports the target verifier at all. ProcessBench itself frames earliest-error identification as a benchmark on which existing PRMs often fail to generalize across sources [7]. Qwen2.5-Math PRMs are strong public baselines trained at much larger scale and provide an external calibration point for our results (§6) [8, 9]. Concurrent work studies PRM transfer across GSM8K, MATH, OlympiadBench, and OmniMath more directly. ProcessLID reports explicit transfer tables across these sources [15], and RetrievalPRM frames PRM failures as question/step out-of-distribution issues across the same sources [16]. Our contribution relative to this thread is not that cross-source ProcessBench transfer is itself novel, but the controlled contrast between generated teacher labels and ProcessBench-style gold labels under one fixed student, optimizer, and evaluation path (§6).

2.4 Expert–amateur contrast

Our framing descends from work that extracts signal from the gap between a stronger and a weaker model. CLEAR [5] contrasts feedback from a larger "expert" and a smaller "amateur" model and applies the refined difference to iteratively improve outputs across reasoning tasks; it is the direct predecessor of the present project. LightReasoner [6] similarly exploits the behavioral divergence between a strong expert and a weak amateur to pinpoint high-value reasoning moments and distill them, notably without ground-truth labels, to sharpen reasoning. We retain the ground-truth-free constraint at student test time but invert the axis of interest. Rather than contrasting models of different capacity, we study a teacher with a privileged view, meaning access to the reference solution while labeling, distilled into a student that is answer-blind by construction. The relevant axis for us is information, not size.

2.5 Data-centric learning and privileged information

Data-centric AI work argues that systematic changes in data quality can dominate model-side changes [17]. Our §6 result is a PRM-specific instance of that pattern: a fixed small verifier's performance changes substantially with the supervision source, holding the model and optimizer fixed. The privileged-label experiments in §4–§5 are also related to learning using privileged information [18], but with an important boundary: privileged information can improve the teacher's per-example inference without producing labels that distill into a better answer-blind student. This distinction is especially important relative to recent on-policy self-distillation methods: OPSD and position-weighted OPSD condition a teacher policy on privileged solution information and distill dense token-level guidance into the same model's policy on its own rollouts [12, 13]. We instead distill off-policy step-error labels from an external teacher into a separate answer-blind verifier, so our negative privileged-label result identifies a failure mode for format-sensitive verifier supervision rather than contradicting successful privileged on-policy policy distillation. Indeed, our diagnosis is that off-policy, format-mismatched supervision fails to transfer. That is consistent with the distribution-matching argument motivating the on-policy line in the first place [14], observed here from the verifier side rather than the policy side.

2.6 Our position

We make three contributions, and we are deliberate about which is load-bearing. First (§4), we characterize *when* privileged supervision is worth anything at the teacher level: a privilege $\times$ difficulty $\times$ richness effect in which a teacher's answer-aware judgment improves step-error localization only when the problem is hard enough that the teacher cannot self-verify *and* the privilege is rich enough to act on. To our knowledge this conditional structure – privilege buys signal precisely where self-verification fails, and only when the privileged information is a reasoning trace rather than a number, has not been isolated in the PRM or expert–amateur literature. This is the validated result and the paper's spine. Second (§5), we show this advantage does not distill into a small student verifier, and we rule out the most obvious confound with a positive control: that the training pipeline simply cannot detect any teacher-quality difference. Third (§6), we diagnose the failure further: under an identical training/evaluation path, format-matched process supervision does produce a competent verifier while raw generated-label rendering does not; the format-rerender cell then identifies label convention as a major bottleneck.

3 Experimental Setup

3.1 Teacher-level privilege probe (§4)

We evaluate a teacher's ability to localize the first erroneous step in a multi-step solution under three supervision conditions that differ only in the privileged information made available to the teacher at labeling time: (i) **no-GT**, an answer-blind condition in which the teacher sees only the problem and the candidate solution; (ii) **+answer**, in which the teacher additionally sees the final reference answer; and (iii) **+solution**, in which the teacher sees the full worked reference solution. Performance is reported as ProcessBench-style first-error F1, together with error-detection accuracy. Reported deltas are taken against the no-GT condition, which serves as the unprivileged reference point.

3.2 Student distillation (§5)

The student is `Qwen2.5-1.5B-Instruct` with a lightweight score-and-critique head, trained via LoRA on teacher-produced score and natural-language critique labels ($N_{\text{train}} = 1000$, $N_{\text{eval}} = 400$, 2 epochs). We compare students trained on privileged (+solution) versus no-GT teacher labels, and additionally a privileged-score-only ablation that drops the critique loss.

3.3 Distribution-matched diagnostic (§6)

To separate student capacity from supervision distribution/format, we run a second, matched training/evaluation path on `Qwen2.5-3B-Instruct` with a score-only head, LoRA adapters, binary cross-entropy loss, balanced batches, and 500 training steps. All headline results in §6 evaluate on the same shuffled ProcessBench-MATH split of 1,000 solutions (6,505 total steps, 594 error steps), using exact serial scoring (batch size 1). We compare two supervision sources under this identical path: (a) ProcessBench-style gold step labels from the first 400 examples of a non-MATH source configuration (GSM8K or OmniMath), flattened to per-step labels, and (b) our generated teacher labels from the privileged/no-GT labeling pipeline described above. Exact problem-string overlap between each source-config training file and the MATH-1000 evaluation split is zero.

4 The Privilege Sweet Spot (Teacher Level)

4.1 Teacher selection

Before probing the effect of privilege, we fix the teacher by an upper-bound bake-off in which every candidate is granted ground-truth access (the with-GT ceiling, $N = 30$). Appendix A.1 shows the official Gemma-class checkpoint attains the strongest first-error F1 (0.909) with perfect format compliance (zero parse failures) and the highest throughput among the candidates; a comparably-sized Qwen variant follows closely (F1 0.873). A reasoning-tuned candidate collapses under the structured-output protocol, failing to parse on the large majority of instances (parse-failure rate 0.846, F1 0.000). Verbal reasoning fluency does not by itself confer reliable structured step adjudication. We adopt the Gemma checkpoint as the primary teacher and retain the Qwen variant for the cross-family replication reported below.

4.2 The sweet spot

Our central finding is that privileged supervision is *conditionally* beneficial: it helps only when the problem is hard enough that the teacher cannot self-verify and tractable enough that the teacher can still exploit the reference. The solution-privilege gap is non-significant on easy problems (GSM8K, -0.052, n.s.),[1] opens to a significant $+0.051$ F1 on hard-but-tractable problems (MATH, 95% CI $[0.01, 0.09]$), and returns to non-significant on the hardest tier (OlympiadBench, -0.03, n.s.). The three points trace an inverted U in difficulty, but we do not lean on the two non-significant endpoints alone to assert a curve; the curvature is corroborated by the mechanism analysis in §4.4, where the rescue rate declines monotonically with reference length. The reading the data support is conjunctive: privilege pays off only where the teacher both needs the reference and can use it.

Table 1 reports the three tiers in full, together with the Qwen-27B cross-family replication. On GSM8K, neither the bare answer nor the full solution improves an already-saturated judgment; the teacher is, in effect,

[1] This near-zero GSM8K gap is measured on the $N = 50$ teacher probe under the first-error F1 protocol used throughout this section. In the larger same-source labeling run of §6.1 ($N = 400$, a different protocol and metric), the privileged teacher does show a $+0.048$ first-error *exact-match* advantage on GSM8K. The two are not in conflict: the sweet-spot claim concerns where the privilege advantage is *large*, and both measurements agree that it is small on GSM8K relative to MATH (+0.10–0.13 on the same scales).

Tier	Benchmark (teacher)	no-GT	+answer	+solution	Δ (solution)
Easy	GSM8K (Gemma, $N = 50$)	0.889	0.856	0.837	-0.052 (n.s.)
Hard	MATH (Gemma, $N = 400$)	0.726	0.726	0.777	$+$**0.051** $[0.01, 0.09]$
Hardest	OlympiadBench (Gemma, OE)	–	–	–	-0.03 (n.s.)
Hard	MATH (Qwen-27B, $N = 150$)	0.653	0.690	0.735	$+0.082$

Table 1: The privilege sweet spot across difficulty tiers, with cross-family replication. Values are first-error F1; Δ is the +solution condition minus the no-GT condition. MATH (Gemma) is the $N = 400$ definitive run (95% CI shown). The OlympiadBench row is evaluated on the open-ended (OE) subset only, under a separate scoring protocol whose per-condition F1 is not on the same scale as the ProcessBench-style rows; we therefore report the paired within-run Δ (-0.03, n.s.) and leave the per-condition cells blank rather than print numbers that invite a cross-row comparison they do not support.

its own oracle, and external privilege is redundant. On MATH, where unaided self-verification degrades, the full worked solution lifts first-error F1 from 0.726 to 0.777 (+0.051, 95% CI $[0.01, 0.09]$, significant) and error recall from 0.608 to 0.661. A smaller preliminary MATH run ($N = 50$) points the same way with a larger nominal gap ($\sim +0.13$ F1). OlympiadBench, harder still, returns the gap to ≈ 0: the references are too long for the teacher to follow, so the privilege it is handed is no longer usable. The same inverted U faintly reappears within MATH by difficulty level (strongest near the middle, weakest at the easiest and hardest levels), corroborating that tractability is the moderator; but the load-bearing evidence is the three-tier contrast, not the noisier per-level split. Critically, the cross-family Qwen-27B teacher reproduces the effect on MATH (+0.082 F1, error detection $0.526 \rightarrow 0.667$), which substantially weakens any single-model-artifact explanation.

4.3 Richness, not mere access

The MATH panel isolates a second, sharper effect. The +answer condition supplies a correct final answer but no reasoning, and it is indistinguishable from the answer-blind baseline: the two rows are identical, and the bare answer flips zero predictions (measured at $N = 150$; we verified the labeling pipeline is correctly wired, so the null effect is behavioral, not an artifact). Only the full worked solution moves the metric. The operative variable is therefore not whether the teacher is privileged, but whether the privilege is *rich*: a reference reasoning trace supplies the step-by-step scaffold needed to localize an error, whereas a final answer, which the teacher cannot back-propagate into the faulty step, supplies nothing it can act on.

4.4 Mechanism: privilege rescues self-verification failures

The sweet spot is not a raw difficulty effect; it is a rescue of self-verification failures, decomposing into two gates ($N = 400$). **Gate 1 (the teacher must need help).** Of 227 problems containing an error, the no-GT teacher misses 89. Adding the worked solution rescues 29 of those (33%), while breaking only 17 of the 138 it had already caught (12%) – the benefit concentrates exactly where self-verification fails. **Gate 2 (the teacher must be able to use it).** The rescue rate falls monotonically with the length of the reference solution (short $0.37 \rightarrow$ mid $0.33 \rightarrow$ long 0.28), which is the mechanism behind the collapse at the hardest tier. Together the gates identify the moderator as self-verification-failure $\times$ tractability, precisely the inverted U observed both across and within datasets.

5 Does the Teacher Advantage Transfer to the Student?

The teacher-level sweet spot in §4 is the validated result. The natural next question is whether it distills: if we train a small (1.5B) student PRM on the privileged teacher's score-plus-critique labels, does that student inherit a usable advantage over one trained on no-GT labels? Across a validated $N = 1000$ run (1,000 training problems, 400 evaluation problems, Gemma-4 labeling confirmed through $\sim$32k served requests), the answer is no. We report this as a clean, diagnosed negative, not a dead end.

Problem (MATH level)	What no-GT missed	+solution feedback at the flagged step
Tens digit of $7! + 8! + \cdots + 2006!$ (L3)	Predicted no error anywhere (missed step 3)	"Incorrectly assumes the tens digit of a sum is simply the sum of the tens digits; fails to account for a carry from the units place."
Regular-tetrahedron centroid ratio (L5)	Flagged step 1 instead of the actual error at step 4	"Incorrectly concludes $PQ/AQ = 1/3$ from a 3:1 median-division claim; in a tetrahedron the centroid divides the altitude so that $AP/PQ = 3{:}1$, not PQ/AQ."
Base-6 representation of 355 (L2)	Predicted no error anywhere (missed step 0)	"The division $355 \div 6$ is incorrect; $355 = 59 \times 6 + 1$, so the remainder should be 1."

Table 2: Three rescue cases from the N=400 evidence pack in the anonymized supplementary material (gold error step known, no-GT prediction wrong, +solution prediction correct), selected to span error types: a missed carry/place-value step, a misapplied geometric ratio, and a missed arithmetic slip. 37 such rescue cases exist in the N=400 evidence pack; these three are illustrative, not cherry-picked for effect size.

Student (teacher labels)	ROC-AUC	PR-AUC	F_1	error recall
Privileged + critique	0.631	0.130	0.170	0.357
Privileged, score-only	0.624	0.135	0.184	0.427
No-GT + critique	**0.641**	0.121	0.198	0.467

Table 3: Step-level transfer ($N_{\text{train}} = 1000$, $N_{\text{eval}} = 400$). The no-GT student is highest on ROC-AUC, so privilege does not transfer.

5.1 Step-level transfer

We compare three students on threshold-free ROC-AUC: privileged+critique, privileged score-only, and no-GT+critique. ROC-AUC is the appropriate measure here because first-error F1 at a fixed score-head cutoff moves with calibration rather than capability. Table 3 shows the no-GT student is highest on ROC-AUC (0.641 vs. 0.631), and the run is non-degenerate (predicted-error rate ≈ 0.3 across all cells, so no silent collapse inflates the comparison). Privilege does not produce a better student verifier.

5.2 Downstream best-of-N

The motivating promise of a ground-truth-free step-error localizer is a test-time gain: given N sampled solutions, the student PRM scores each and we re-rank, selecting the highest-scoring candidate. Table 4 reports the $N = 1000$ result. The no-GT verifier re-ranks *better* than the privileged one (0.373 vs. 0.349), and decisively neither beats the majority-vote baseline (≈ 0.39).[2] The oracle (0.517) shows substantial headroom remains, so the failure is the verifiers', not the candidate pool's.

5.3 Why the null: a diffuse, not absent, teacher gap

The negative is not that privileged and no-GT labels are identical. A probe of the actual Gemma-4 labeling teacher reproduces a real +0.07 solution gap (F1 0.716 → 0.786; error recall 0.577 → 0.654, $N = 150$ MATH), consistent with §4. The issue is the *shape* of the gap. Over 6,340 labeled steps, privileged and

[2] Grading protocol caveat: the answer-equivalence check behind Table 4 fell back to string matching rather than symbolic verification, which understates absolute accuracy on MATH answers with non-canonical but equivalent forms. Because the fallback applies identically to every column, since the same graded pool underlies pass@1, PRM re-rank, majority vote, and the oracle, the *comparisons* reported here (no-GT ≥ privileged; neither ≥ majority vote) are unaffected in direction, but the absolute values should be read as lower bounds. A symbolic regrade of the same cached candidate pools is in progress and will be reported in the camera-ready.

Verifier	pass@1	prm rerank	majority vote	oracle pass@N
No-GT student	0.337	**0.373**	0.391	0.517
Privileged student	0.335	0.349	0.382	0.518

Table 4: Downstream best-of-N re-ranking ($N = 1000$ problems, 8 candidates, `gemma-2-9b-it` generator). The no-GT verifier re-ranks better; neither beats majority vote.

no-GT labels agree on only 69%; on the $\sim 31\%$ where they differ, the disagreement is nearly symmetric: privilege flags an error where no-GT does not on 1,001 steps, while no-GT flags one where privilege does not on 956. The teacher's privilege churns labels in both directions; its small net quality gain is diffuse, not a clean directional signal a 1.5B student can latch onto. A same-pool paired best-of-N confirms the consequence: the privileged and no-GT re-rankers are statistically indistinguishable (McNemar $p = 0.14$; 5 vs. 12 discordant pairs), both below majority vote. This $N = 200$ test is underpowered (17 discordant pairs); a properly powered shared-pool $N = 1000$ paired comparison is left to future work, and we scope the claim accordingly: the paired evidence shows no detectable difference, and the unpaired $N = 1000$ evidence (Table 4) points the same way.

The same-source control in §6.1 reproduces this dissociation under tighter conditions, and is worth stating here because it is the cleanest version of the effect. On the same 400 GSM8K problems, the privileged teacher's labels are the more accurate ones *at the teacher level* (first-error exact match 0.800 vs. 0.752), yet the student trained on them is significantly *weaker* than the no-GT-trained student (0.5494 vs. 0.6183 mean ROC-AUC; paired sequence-cluster bootstrap gap +0.0242, 95% CI $[0.0135, 0.0344]$, $p = 0.0004$). A better labeler does not imply a better student: the teacher-level and student-level orderings can invert even with source problems held fixed.

5.4 Positive control: the pipeline is sensitive to teacher quality

The two results above leave one obvious confound open: does the training pipeline fail to distill privilege specifically, or is it simply insensitive to *any* teacher-quality difference? We test this directly. We relabel the same candidate solutions with a deliberately weaker labeling model (`Qwen2.5-3B-Instruct`, in place of the 26B Gemma-4 teacher) and train an identically-configured 1.5B student on the resulting labels. This weak-teacher student reaches ROC-AUC 0.575 on the full 1,000-problem ProcessBench-MATH split (6,505 steps), compared to 0.625 for the Gemma-4-labeled privileged+critique student evaluated on the identical 6,505-step split.[3] A paired bootstrap (10,000 resamples, same procedure as §6.1) on this ROC-AUC gap, recomputed from the currently-committed per-step scores, gives a mean difference of +0.0497, 95% CI $[0.0206, 0.0783]$, two-sided $p = 0.0012$, significant and excluding zero. The weak-teacher student's score head is also markedly less calibrated (it predicts "error" on only 1.6% of steps, versus $\approx 30\%$ for the Gemma-4-labeled cells), so part of this gap reflects outright degeneracy rather than a smooth quality gradient; we report the comparison as-is rather than adjusting for it.

This is the paper's key validity result for §5: because student quality tracks teacher quality when the teacher-quality difference is large and unprivileged (weak vs. strong labeling model), the training pipeline is demonstrably capable of transmitting a teacher-quality signal into the student. The null in §5.3 is therefore not explained by pipeline insensitivity. It is specifically the privileged-vs-no-GT distinction that fails to distill at this student scale and training recipe. That signal is much subtler and more diffuse than a weak-vs-strong difference in teacher quality. This motivates the next question directly: is the remaining gap a matter of student capacity, or of the generated-label supervision itself? We take this up in §6.

[3] This 0.625 differs slightly from the 0.631 in Table 3, which is an earlier 400-problem-eval snapshot of the same condition; the checkpoint was subsequently re-evaluated at full N=1000. To avoid any ambiguity from intermediate retrains, the comparison in this section is recomputed directly from the currently-committed per-step score files for both conditions, so it is independently reproducible from the released artifacts as they stand.

Training source	Seeds	ROC-AUC	PR-AUC	Best F_1^*	Fixed F_1	Pred error rate
GSM8K gold → MATH1000	0–3	0.7515 (0.7256–0.7760)	0.2188 (0.1935–0.2317)	0.3144	0.2503	0.5111
OmniMath gold → MATH1000	0–3	0.7694 (0.7539–0.7869)	0.2524 (0.2277–0.2877)	0.3276	0.3055	0.2920
Qwen2.5-Math-7B-PRM800K (public)	–	0.8379	0.3254	0.3991	0.3953	0.1436
Generated privileged labels, BCE	0–3	0.5475 (0.5007–0.5883)	0.0966 (0.0881–0.1045)	0.1853	0.1089	0.2903
Generated no-GT labels, rank loss	0–3	0.6062 (0.5390–0.6755)	0.1254 (0.0911–0.1672)	0.2142	0.1918	0.7341
Same-source GSM8K generated priv BCE	0–3	0.5494 (0.4680–0.6371)	0.0985 (0.0809–0.1235)	0.1931	0.1741	0.4696
Same-source GSM8K generated no-GT BCE	0–3	0.6183 (0.5849–0.6614)	0.1152 (0.1049–0.1314)	0.2168	0.2088	0.4629
Same-source GSM8K generated priv BCE, PB-format	0–3	0.6762 (0.5869–0.7891)	0.1761 (0.1175–0.2811)	0.2554	0.2385	0.5121
Same-source GSM8K generated no-GT BCE, PB-format	0–3	0.7366 (0.7011–0.7555)	0.2478 (0.2266–0.2722)	0.3215	0.3120	0.1703

Table 5: Full ProcessBench-MATH ($N = 1000$) transfer results under an identical Qwen2.5-3B score-head training/eval path. PB-format rows re-render the same Gemma-4 labels into the ProcessBench first-error convention. Parentheses show seed ranges. *Best F_1 is threshold-swept on the eval slice; treat as diagnostic (Appendix A.4 reports held-out calibrated F_1). Qwen2.5-Math-7B-PRM800K is an external calibration baseline, not a matched training recipe. Our result is not a SOTA PRM claim.

6 Diagnosing the Null: Capacity or Supervision?

Section 5 leaves a specific question open. Having ruled out pipeline insensitivity (§5.4), is the failure of privilege to distill because the 1.5B student lacks the capacity to represent the signal, or because the generated-label supervision itself is poorly matched to what the ProcessBench-MATH evaluation needs? We address this with a controlled ablation on `Qwen2.5-3B-Instruct` (§3): holding the verifier architecture, training path, and evaluation benchmark fixed, we vary only the source and format of step-level supervision.

6.1 Format-matched labels transfer; raw generated labels remain weak

Table 5 shows the main result. With the student, training path, and evaluation held fixed, ProcessBench-style labels from GSM8K and OmniMath transfer strongly to full ProcessBench-MATH. Generated teacher labels are much weaker in their raw rendering, even when the generated-label baseline is chosen as the strongest checkpoint from a broader preliminary sweep (§6.4). Every GSM8K and OmniMath seed significantly beats both raw generated-label baselines under sequence-cluster bootstrap (resampling whole solutions rather than individual steps, which are correlated); against the best generated-label baseline overall, gaps range from +0.093 to +0.155 ROC-AUC across the eight seeds, all two-sided $p = 0.0010$ (Appendix A.2). Two generated-label quantities appear in this section and should not be confused: the *4-seed aggregates* in Table 5 (privileged BCE 0.5475, no-GT rank 0.6062), and the single strongest generated-label *checkpoint* from the preliminary sweep (no-GT rank loss, seed 0, ROC-AUC 0.6324), which we use as the conservative comparison point in the bootstrap and calibration tables because it is the hardest generated-label target to beat.[4] In the same-source GSM8K control, even the weakest GSM8K gold seed beats the best same-source generated no-GT seed by +0.064 ROC-AUC (95% CI $[0.045, 0.083]$, two-sided $p = 0.0004$), so the generated-label weakness is not explained by source distribution alone. The follow-up format-rerender cell changes the diagnosis: using the same Gemma-4 labels but the ProcessBench first-error convention raises same-source no-GT to 0.7366 mean ROC-AUC and same-source privileged to 0.6762. The no-GT 4-seed mean-score artifact reaches 0.7599 ROC-AUC and the privileged 4-seed mean-score artifact reaches 0.7789, statistically tied with the GSM8K gold mean-score artifact under sequence-cluster bootstrap.

Held-out threshold calibration (threshold chosen on the first 200 MATH sequences, evaluated on the remaining 800) agrees with the ranking result: GSM8K and OmniMath gold seeds reach calibrated F_1 of 0.27–0.33, against 0.17 for generated privileged BCE and 0.21 for the best generated-label baseline (Appendix A.4).

[4] Run-to-run variance on this track is substantial and we disclose it rather than aggregate it away: re-running the privileged BCE seed-0 configuration from scratch produced 0.477 ROC-AUC against a canonical 0.550, with both runs collapsing to a predicted-error rate of 0. The seed ranges in Table 5 therefore reflect genuine training instability on generated labels, not measurement noise, and are part of what a format-matched pre-check buys you: the gold-label and PB-format rows show no comparable degeneracy.

6.2 The source distribution still matters

This is not a blanket claim that any non-MATH process labels transfer cleanly to MATH. OlympiadBench is high-variance in our runs: two seeds give 0.585 and 0.716 ROC-AUC on full MATH-1000 (mean 0.651), one significantly below the best generated-label baseline (-0.047 ROC-AUC, $p = 0.0010$) and the other significantly above it ($+0.084$, $p = 0.0010$). We treat OlympiadBench as a boundary diagnostic rather than a third clean headline source; the pattern still supports a source-distribution view, but the source split matters.

6.3 The failure is not explained by the score-head machinery alone

A plausible concern is that the 3B score-head architecture is simply too weak to learn step-level mathematical verification. Appendix A.3 argues against this as the primary explanation. Training the same score-head path on the first 200 shuffled MATH examples and evaluating on the next 200 reaches 0.7129 ROC-AUC. This is not a benchmark claim, since train and eval come from the same benchmark family, but it is evidence that the machinery can learn the target when labels and distribution are aligned. Frozen-representation probes tell the same story: class-balanced logistic probes on frozen Qwen2.5-3B step-boundary embeddings trained on generated teacher labels transfer weakly (0.52–0.56 ROC-AUC), while the matched ProcessBench-gold probe reaches 0.631. The failure is not merely that the student representation cannot expose the target; the supervision distribution is doing meaningful work.

6.4 Training-recipe changes did not rescue generated labels

Before reframing the result as a supervision mismatch, we tested a range of score-head and distillation variants on the generated labels (Appendix A.5). Naive score-plus-critique distillation was unstable until a feedback-token truncation bug was fixed; after the fix, equal-weight critique distillation remained weak or negative for privileged labels. Pure ranking loss and BCE-plus-rank did not rescue privileged transfer, and often favored the no-ground-truth labels instead. Longer BCE training increased variance without turning the generated privileged labels into a competent MATH verifier. These negative ablations do not prove no generated-label method can work, but combined with the 1.5B ablations in §5 (which tried soft-BCE, hard-verdict, and logit-KD distillation with the same result), they reduce the likelihood that the observed gap is a single bad optimizer setting at a single student scale.

6.5 Interpretation

Taken together with §5, the picture is: the training pipeline is sensitive to teacher quality (§5.4); the score-head machinery and student representation can learn a competent verifier when supervision is distribution-matched (§6.1–6.3); and a broad sweep of loss functions and student scales fails to rescue the generated-label signal specifically (§5, §6.4). The most parsimonious explanation across all three tracks of evidence is a supervision rendering/convention mismatch in the raw generated-label pipeline, not a capacity ceiling in the student. The format-rerender cell is the mechanism check: when source problems and teacher labels are held fixed but rendered in the ProcessBench convention, much of the transfer gap disappears.

7 Limitations

Confounded comparison in §6. The gold-vs-generated contrast varies provenance (human-curated ProcessBench annotations vs. LLM-generated labels), source-problem distribution, and label format simultaneously. The same-source GSM8K control removes the source-distribution confound for one source split, and the format-rerender cell shows that prompt/rendering convention explains much of the raw generated-label failure. It still should not be read as fully isolating human-curated versus generated provenance: the rerendered teacher first-error position still disagrees with gold on roughly a quarter of source solutions, so label content and provenance remain coupled.

Training instability and scale asymmetry. The seed asymmetry that originally qualified this comparison is now largely closed: the headline generated-label rows in Table 5 are four-seed aggregates, matching

Teacher	F_1	correct acc	error acc	parse fail
Gemma-4-26B (primary)	**0.909**	1.000	0.833	0.000
Qwen-27B	0.873	0.917	0.833	0.000
Qwen-35B-A3B	0.839	1.000	0.722	0.019
Qwen-4B-Thinking	0.000	1.000	0.000	0.846

Table 6: Teacher bake-off, with-GT ceiling ($N = 30$).

the gold-source rows, and the gap survives at every seed. What replaces it is a narrower caveat about variance. Generated-label training is markedly less stable than gold-label training. Seed ranges span $\sim$0.09 ROC-AUC for privileged BCE and $\sim$0.14 for no-GT rank loss, against $\sim$0.05 and $\sim$0.03 for the GSM8K and OmniMath gold rows, and some runs collapse to a degenerate all-negative score head. We read this instability as part of the finding rather than as noise to be averaged out, but it does mean the generated-label means are less well determined than the gold-label means. Separately, the 1.5B (§5) and 3B (§6) tracks are not a controlled capacity ablation of one another, since they differ in loss recipe and evaluation N. The observation that generated-label transfer stays weak at both scales is therefore suggestive, not a clean single-variable ablation of student capacity.

Not a SOTA PRM claim. The public Qwen2.5-Math-7B-PRM800K baseline is stronger than any single gold-source seed on the same split (Table 5). The contribution is diagnostic: a format-matched cheap check can reveal supervision mismatch before investing in a larger student or a broader recipe search, not a new state-of-the-art verifier.

OlympiadBench and the positive control both carry caveats. OlympiadBench is high-variance over two seeds and is treated as a boundary diagnostic, not a headline source. The positive control in §5.4 uses a weak-teacher student whose score head is also miscalibrated (near-zero error-prediction rate), so part of the measured gap may reflect outright degeneracy rather than a smooth teacher-quality gradient.

Downstream validation of §6 is incomplete. Unlike §5, the 3B gold-vs-generated result is evaluated only on ProcessBench step-level metrics; we do not show the gold-supervised 3B verifier beats majority vote or the generated-label student on downstream best-of-N re-ranking.

8 Conclusion

Under a fixed 3-tier teacher probe, privileged (answer-aware) process supervision helps a teacher precisely in a tractability window, where it is both needed and usable, and this effect replicates across model families. That advantage does not distill into a small student verifier: a 1.5B student trained on privileged labels is statistically indistinguishable from one trained on no-GT labels, and neither clears majority vote on downstream best-of-N. This null is not explained by an insensitive training pipeline, which we confirm with a positive control. Diagnosing further, we show that under an identical training/evaluation path, format-matched gold process supervision produces a competent verifier while our generated teacher labels do not, at both 1.5B and 3B student scale. This points at supervision distribution and format mismatch, not student capacity, as the dominant explanation. For efficient PRM development, this suggests a practical ordering: before scaling students or re-running the same distillation loop, test whether the supervision distribution itself can support the target verifier.

A Supplementary Tables

Tables moved out of the main text for length. All numbers are identical to those cited in the body and trace to committed artifacts under `results/`.

Model A	Model B	ROC-AUC gap	95% CI (p)
GSM8K gold seed 0	best generated-label baseline	+0.111	$[0.087, 0.134]$ (0.0010)
OmniMath gold seed 0	best generated-label baseline	+0.121	$[0.096, 0.146]$ (0.0010)
GSM8K gold seed 0	generated privileged BCE	+0.193	$[0.164, 0.221]$ (0.0010)
OmniMath gold seed 0	generated privileged BCE	+0.203	$[0.176, 0.232]$ (0.0010)
Qwen PRM800K	OmniMath gold seed 3 (best single seed)	+0.051	$[0.033, 0.068]$ (0.0010)
PB-format no-GT 4-seed mean	raw no-GT 4-seed mean	+0.126	$[0.104, 0.148]$ (0.0004)
PB-format priv 4-seed mean	raw priv 4-seed mean	+0.207	$[0.183, 0.231]$ (0.0004)
GSM8K gold 4-seed mean	PB-format priv 4-seed mean	+0.004	$[-0.008, 0.017]$ (0.4983)

Table 7: Representative sequence-cluster bootstrap ROC-AUC gaps. Older generated-baseline rows use 2,000 resamples; same-source and format-rerender rows use 5,000 resamples. Full bootstrap tables for every pairwise comparison are included in the anonymized supplementary material.

Diagnostic	Condition	ROC-AUC
Score head, matched ProcessBench-MATH 200/200	Gold labels	0.7129
Frozen probe, generated teacher labels	Privileged	0.5497
Frozen probe, generated teacher labels	No-GT	0.5559
Frozen probe, ProcessBench-gold 200/200	Gold labels	0.6308

Table 8: Diagnostics separating score-head/representation capacity from supervision mismatch.

A.1 Teacher selection

A.2 Sequence-cluster bootstrap gaps

A.3 Capacity-versus-supervision diagnostics

A.4 Held-out threshold calibration

A.5 Generated-label training-recipe ablations

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Model	Calibrated F_1	Eval ROC-AUC	Eval PR-AUC	Pred error rate
GSM8K gold seed 0	0.3166	0.7452	0.2339	0.2488
OmniMath gold seed 0	0.3062	0.7483	0.2292	0.2742
Qwen2.5-Math-7B-PRM800K	0.3855	0.8372	0.3242	0.1091
Generated privileged BCE	0.1673	0.5490	0.1002	0.3212
Best generated-label baseline	0.2085	0.6314	0.1431	0.1695

Table 9: Held-out threshold calibration (threshold on first 200 MATH sequences, evaluated on remaining 800).

Generated-label recipe	Priv ROC-AUC	No-GT ROC-AUC	Note
Score+critique MSE, early run	0.4582	0.4870	collapsed near all-negative
Score-only MSE	0.4844	0.4776	stable but weak
Feedback-token truncation fixed	0.4514	0.5093	stable but negative for priv.
BCE, error weight 3	0.5411	0.4700	best short privileged ablation, still weak
Rank-only, balanced batches	0.4939	0.6306	no-GT wins strongly
BCE+rank, balanced batches	0.5214	0.6140	no-GT wins strongly
BCE, error weight 3, longer training	0.4782	0.4971	longer BCE degrades/nulls signal

Table 10: Preliminary generated-label ablations on the Qwen2.5-3B score-head path. Representative rows from a ∼10-configuration sweep; no variant rescues privileged-label transfer.

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